

Project - Retail Sales Data

1. Project Overview :-

This dataset captures retail transactions across multiple regions (East, West, North, South) with details on:

- Customers
- Product categories (Clothing, Electronics, Home, Sports)
- Order details (quantity, unit price, order date)
- Payment methods (Cash, Credit Card, UPI, Netbanking)

The goal is to answer client-style business questions such as revenue analysis, customer ranking, regional comparisons, and monthly sales trends.

2. Data Structure :-

Columns :-

- OrderID → Unique identifier for each transaction
- CustomerID → Unique customer reference
- Region → Geographic region of sale
- ProductCategory → Category (Clothing, Electronics, Home, Sports)
- ProductName → Individual product identifier
- Quantity → Units sold
- UnitPrice → Price per unit
- OrderDate → Date of transaction
- PaymentMethod → Mode of payment

3. Derived column to create :-

- Revenue = Quantity × UnitPrice
- OrderValue = Revenue per transaction

4. Key Insights :-

- Electronics often have high unit prices (e.g., Product_6 at ₹815, Product_91 at ₹998).
- Clothing shows wide variation: some low-priced (₹120) and some premium (₹927).
- Sports products frequently have bulk quantities (7–9 units), driving higher order values.
- Payment methods are evenly spread, but Cash and Netbanking dominate large orders.

- Seasonal spikes: December shows heavy sales (holiday effect).
- Observed Patterns
- Regional differences :-
 - East & West regions show strong Clothing and Electronics sales.
 - South region has balanced sales across all categories.
 - North region leans toward Home and Sports.
- Customer concentration: Some customers (e.g., IDs 1957, 1339, 1455) appear multiple times, suggesting repeat buyers.
- Monthly trends: Peaks around March, June, September, December → possible seasonal demand cycles.

❖ Client Questions :-

1. Which product category generated the highest revenue last quarter?

Solution :-

SQL

- a. First add column Total plus order date in only month format
- b. Use SUM(Total) grouped by ProductCategory.
- c. Use where clause and between to find last 3 months revenue data
- d. GROUP BY Region for average order value.
- e. ORDER BY revenue descending for top customers.

	A	B	C	D	E	F	G	H	I	J
1	Order_id	Customer	Region	Product_c	Product	Quantity	Unit_price	Order_date	Payment_method	
2	1	1182	East	Clothing	Product_0	7	309	*****	Credit Card	
3	2	1435	West	Clothing	Product_1	4	405	*****	Upi	
4	3	1860	South	Home	Product_2	6	369	*****	Netbanking	
5	4	1270	West	Sports	Product_3	9	803	*****	Cash	
6	5	1106	South	Electronic	Product_4	1	786	*****	Cash	
7	6	1071	West	Home	Product_5	5	354	*****	Upi	
8	7	1700	East	Electronic	Product_6	7	815	*****	Upi	
9	8	1020	South	Clothing	Product_7	1	775	*****	Netbanking	
10	9	1614	East	Clothing	Product_8	4	862	*****	Cash	
11	10	1121	South	Sports	Product_9	6	326	*****	Cash	
12	11	1466	East	Clothing	Product_1	2	480	*****	Cash	
13	12	1214	East	Clothing	Product_1	5	559	*****	Cash	
14	13	1330	South	Home	Product_1	1	106	*****	Cash	
15	14	1458	South	Electronic	Product_1	6	128	*****	Credit Card	
16	15	1087	South	Electronic	Product_1	6	132	*****	Upi	
17	16	1372	North	Home	Product_1	4	745	*****	Credit Card	
18	17	1099	West	Sports	Product_1	8	978	*****	Cash	
19	18	1871	North	Electronic	Product_1	4	178	*****	Netbanking	
20	19	1663	North	Clothing	Product_1	8	558	*****	Netbanking	
21	20	1130	East	Electronic	Product_1	8	669	*****	Cash	
22	21	1861	South	Clothing	Product_2	5	637	*****	Cash	
23	22	1308	West	Sports	Product_2	4	581	*****	Netbanking	
24	23	1769	East	Home	Product_2	2	529	*****	Credit Card	
25	24	1343	North	Electronic	Product_2	3	837	*****	Credit Card	
26	25	1491	South	Home	Product_2	3	384	*****	Cash	
27	26	1413	East	Electronic	Product_2	4	615	*****	Credit Card	
28	27	1805	South	Clothing	Product_2	4	190	*****	Netbanking	
29	28	1385	East	Home	Product_2	5	496	*****	Credit Card	
30	29	1191	South	Electronic	Product_2	4	588	*****	Cash	
31	30	1955	North	Clothing	Product_2	8	138	*****	Netbanking	

SQL File 8* SQL File 7* SQL File 7* retail_sales

Limit to 1000 rows

```

1 • SELECT * FROM df.retail_sales;
2 • select product_category , sum(quantity*unit_price) as Total from retail_sales group by product_category order by total desc ;

```

Result Grid

product_category	Total
Electronics	401103
Home	348000
Clothing	324255
Sports	305076

- Identify the top 5 customers by total purchase value.

Solution :-

Python

- First import csv file in python jupyter notebook with the help of pandas library .
- Extract columns customer_id , quantity and unit_price .
- Create new column in dataset with name Total .
- Now last step do grouping by customer_id with total and export into power bi .

```

111 import pandas as pd
112 df = pd.read_csv('C:/Users/pavankrishna/Desktop/quantity_2.xlsx')
113 df
114
115 Order id Customer id Region Product category Product name Quantity Unit price Order date Payment method
0 1 1102 East Clothing Product_1 7 899 11-12-2024 Credit Card
1 2 1405 West Clothing Product_1 4 455 11-12-2024 Net
2 3 1800 South Home Product_2 6 889 14-12-2024 Netbanking
3 4 1275 West Sports Product_3 9 889 14-12-2024 Cash
4 5 1104 South Electronics Product_3 1 799 14-12-2024 Cash
... ..
492 499 1017 North Home Product_499 1 794 10-10-2024 Net
493 497 1792 North Home Product_498 4 512 11-12-2024 Netbanking
494 495 1754 South Electronics Product_497 1 814 14-12-2024 Cash
495 499 1545 East Electronics Product_496 4 454 10-10-2024 Cash
496 500 1605 North Sports Product_495 8 855 10-10-2024 Net
500 rows x 8 columns

116 df[['Customer_id', 'Quantity', 'Unit_price']]
117 df
118
119 Customer id Quantity Unit price
0 1102 7 899
1 1405 4 455
2 1800 6 889
3 1275 9 889
4 1104 1 799
... ..
492 1017 1 794
493 1792 4 512
494 1754 1 814
495 1545 4 454
496 1605 8 855
500 rows x 3 columns

120 df[['Customer_id', 'Quantity', 'Unit_price', 'Total']]
121 df
122
123 Customer id Quantity Unit price Total
0 1102 7 899 6293
1 1405 4 455 1820
2 1800 6 889 5334
3 1275 9 889 10221
4 1104 1 799 799
... ..
492 1017 1 794 794
493 1792 4 512 2048
494 1754 1 814 814
495 1545 4 454 1816
496 1605 8 855 6840
500 rows x 4 columns

124 df.groupby('Customer_id').sum().reset_index()
125 df
126
127 Customer id Total
0 1102 6293
1 1405 1820
2 1800 5334
3 1275 10221
4 1104 799
... ..
492 1017 794
493 1792 2048
494 1754 814
495 1545 1816
496 1605 6840
500 rows x 2 columns

128 df.to_csv('C:/Users/pavankrishna/Desktop/quantity_2.xlsx')
129 print('Success')
130
Success

```

- Compare average order value across different regions.

Solution :-

Excel

- First open csv file in Excel .
- Now create pivot table according to region wise average of quantity .

Row Labels	Average of Quantity
East	5.351851852
North	4.885714286
South	4.691666667
West	5.136363636
Grand Total	5.006

4. Find monthly sales trends for the last 12 months.

5. **Solution :-**

My Sql

- Open my sql again and create a query with subquery
- Use group by function on order_date plus also aggregate on quantity * unit_price .
- Now export file in csv and load in power bi software .

```
1 • SELECT * FROM df.retail_sales;
2 • select product_category , sum(quantity*unit_price) as Total from retail_sales group by product_category order by total d
3 • select Order_Date , sum(quantity*unit_price) as Revenue from retail_sales group by order_date order by order_date asc ;
```

Order_Date	Revenue
01-02-2024	7615
01-03-2024	2770
01-04-2024	1368
01-05-2024	7320
01-06-2024	3834
01-07-2024	867
01-08-2024	7623
01-09-2024	786
01-10-2024	5448
01-11-2024	1540
01-12-2024	16764
02-01-2024	2115
02-02-2024	5179
02-03-2024	13120
02-05-2024	2728
02-06-2024	5180
02-07-2024	9527
02-08-2024	4270
02-09-2024	6493
02-10-2024	4709
02-11-2024	6342
02-12-2024	8910
03-01-2024	1530
03-02-2024	7168

POWER BI REPORT :-



